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Vowel Quality of Swiss EFL Speakers

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Introduction

The moment we hear someone converse in a foreign language, we are likely to make judgments, conscious or unconscious, about whether this person is a native or non-native speaker.¹ On a linguistic level, we draw such conclusions from a speaker's global foreign accent, which Major (2001, 19) defines as "the overall impression concerning a speaker's form whether or not and to what degree a person sounds native or non-native". The word "accent" is a blurry term in need of further qualification. On the one hand, it can denote the manner of pronouncing the words of a language in such a way that the speaker's regional provenance permeates. On the other hand, accent also refers to the phenomenon of stress prominence in words and phrases. Moreover, there is a third definition of accent – most pertinent to this study – namely that of accent in the context of second language acquisition. This second language (L2) accent is what interests me in particular.

Lippi-Green describes the L2 accent as "breakthrough of native language phonology into the target language" (1997, 43). In other words, the L2 spoken by the non-native speaker thus neither pertains to the L1 nor to the L2. Selinker (1972) captures this phenomenon adequately in coining the term *interlanguage*, which refers to the systematic knowledge of an L2 that is independent of both the target language and the learner's L1. The *interlanguage* can of course be studied on various linguistic levels such as its syntax, lexicon, its use in discourse, suprasegmentally etc. In the case of this study, the focus is put on an acoustic observation of interlanguage, in particular, the interlanguage of Swiss English (henceforth SE), which is submitted to an acoustic phonetic observation.

In Switzerland, where 4 languages are spoken officially and English is continuously gaining impetus, language practice, language use, and language attitudes are often the subjects of heated debates. These days it is especially the issue of whether English should be taught as the first foreign language at primary schools in the German, French, and Italian parts of the country, which is discussed time and again. From personal encounters and observations, I am also under the impression that the pronunciation with which Swiss speakers converse in a foreign language is, too, regularly debated and discussed.² For this reason, it is the accented speech of learners of English as a foreign language (EFL) in Switzerland that is investigated in this study. To be more precise, I will examine the vowel quality of Swiss EFL speakers.

¹ Note: The present paper rests on the author's MA Thesis (Leemann 2006).

² Clearly, this foreign language is assessed by a standard language or standard dialect, i.e. a norm. I will address the concept of Standard Language and its relevance for this study in the following "Key concepts" section.

This paper is subdivided into four main parts. To begin with, I will qualify and define two significant concepts of the study, which are vowel quality, i.e. the reasoning behind why vowel quality was chosen as indicative of a foreign accent, and the comparison to the standard. Then, the method will be addressed, that is the criteria according to which the sample was chosen, reading tasks, questionnaire, recordings, and actual measurement of vowel quality. The main section consists of the results section, which is followed by the final section, designed to seek explanations regarding the obtained results. I will end this paper with a few conclusive remarks and with a brief outlook.

1. Key concepts

The rationale behind this second introductory section is primarily to qualify the object under scrutiny, i.e. vowel quality of Swiss EFL speakers. Secondly, I would like to devote a few words to the issue of Standard Language, i.e. a norm which acts as a point of reference and comparison regarding the performance of non-native speakers' speech, and how this concept is understood in the present study.

1.1 Vowel quality

A foreign accent is acoustically perceptible on different levels. It manifests itself on the segmental and suprasegmental level, of which the former one has been studied meticulously in terms of interlanguage phonologies. For this reason, the present paper argues along similar lines and attempts to highlight how the monophthong vowels as spoken by Swiss English speakers deviate from the vowels as represented in Received Pronunciation (RP) and General American (GA). Primarily, monophthong vowels, as opposed to consonants, were selected because vowels offer a larger "corroding surface" for individual variation; secondly, vowels are not as categorical as consonants. Each vowel bears a distinctive acoustic quality, which can be measured with an adequate signal analysis program. In the case of the present study PRAAT (2008) was used.

Vowel quality is largely determined by the constellation of the first 2 formants, F1 and F2. The lower the F1 value, the closer the tongue to the roof of the mouth, i.e. the higher the vowel. The F2 value, conversely, is proportional to the frontedness or backness of the highest part of the tongue during vowel production. That is to say, the higher the F2 value, the more fronted the vowel. The Swiss English vowel quality is compared to the vowel quality of RP and GA. The values for these two varieties were retrieved from two studies, which will be briefly introduced in the following section.

1.2 Comparison to norm

As mentioned earlier, when we make conscious or unconscious judgments of non-native speakers' speech we do so by comparing it to a norm. Clearly, this raises issues regarding the actual existence of a standard language or a standard dialect. As mentioned earlier, this paper, too, applies a comparison of an interlanguage variety to a standard variety. In the context of this paper, however, standard language is not thought of as a concrete language; rather, the author adheres to Milroy and Milroy's definition of Standard English as "an idea in the mind rather than reality – a set of abstract norms to which actual usage may conform to a greater or lesser extent" (1991, 22-23).

The formant values were taken from two studies, one conducted on General American, the other on Received Pronunciation. The formant values from GA were retrieved from Hillenbrand et al.'s study "Acoustic characteristics of American English vowels" (1995). It was their purpose to replicate a seminal study by Peterson and Barney who explored the acoustic characteristics of American English vowels in the mid 50s (1952). Hillenbrand et al. selected 139 subjects which primarily came from Michigan's Lower Peninsula. The subjects performed reading list tasks which included 12 English vowels in /hVd/ phonological environments.

The values for the RP vowels were taken from Deterding's study "The formants of monophthong vowels in Standard Southern British English pronunciation" (1997). In this study, which retrieved its data from the MARSEC database,³ Deterding investigates the formant of eleven monophthong vowels of Standard Southern British English. Unlike Hillenbrand's study and the present study, which both adhered to prepared speech obtained through word list tasks, Deterding analyzed BBC monologues from the 1980s, which included news readings and commentaries.

At this point it is crucial to reiterate that the two studies are different in terms of how the data was obtained. While Hillenbrand et al.'s (1995) study considered English monophthong vowels in an identical /hVd/ environment, Deterding (1997) studied prepared speech also, yet the target vowels did not occur in identical phonological environments. This difference has repercussions in terms of the vowel quality obtained, where particularly Deterding's (1997) corpus contains fast speech, which in turn shows vowel centralizations and varying coarticulation phenomena (Strik et al. 2000). These differences in data collection and the repercussions thereof shall be borne in mind when the results of the present study are presented.

³ MARSEC stands for *Machine Readable Spoken English Corpus*.

2. Method

The subjects in the study at hand were screened according to their L1 background, L2 experience, age, and gender. One of the most suitable ways to control such variables is in the school context, as variables such as age, sex, L1, and L2 background can be monitored. The secondary school in Zofingen was chosen as recording location due to Zofingen's preferable geographical location, nestled in a valley between the largest Swiss German-speaking cities, Zurich, Bern, Basel, Lucerne; cities which all feature distinctive Swiss German dialects. From a pool of potential 40 subjects, 20 subjects were recorded whose data served as the basis for the corpus of the present study. The students were asked to perform word list reading tasks and fill out a questionnaire – the latter elicited external variables that are known to have an effect on L2 articulation. The following section attends to the reading tasks, the recordings, and the vowel quality measurement applied in this study.

2.1 Word lists

In acoustic phonetics, word lists have become well-established. On the one hand, they are practical in that word lists allow for an integration of vowels in identical environments. Furthermore, the researcher can elicit much data with little effort during the recordings, as word lists can be applied mechanically. The word lists which were applied contained the English target vowels in an /hVd/ phonological environment.⁴ Table 1 shows the word list containing the Swiss English target vowels.

/hVd/ construction	Supporting word	Target vowel
heed	(need)	/i:/
hid	(bit)	/ɪ/
head	(bed)	/ɛ/
had	(sad)	/æ/
hod	(God)	/ɑ:/
hawd	(saw)	/ɔ:/
hood	(good)	/ʊ/
who'd	(noon)	/u:/
hud	(but)	/ʌ/
heard	(bird)	/ɜ/
heed	(need)	/i:/

Table 2-1: Swiss English target vowels.

⁴ The vowels were selected according to Ladefoged's (2001, 27) conception of English vowels.

The word lists are structured according to a number of criteria. A second and third column, not depicted in Table 2-1, repeat the words that appear in the first column, yet in random order, so as to prevent memory effects on the part of the subjects. Furthermore, each column ends with a dummy word, usually the first word in the specific column, as the reading of word lists triggers a falling intonation in the last word, which may affect vowel length and vowel quality. The dummy words are not included in the analyses. The words in brackets – captured as “supporting words” in Table 2-1 - are meant to help the students pronounce the target words correctly. By means of a questionnaire, a number of factors which are considered to affect L2 articulation were elicited – more on these external variables in the results section.

2.2 Recordings and measurement

Recordings were made with four Digital Audio Tape (DAT) recorders⁵ and signal analysis was performed with PRAAT (2008). The DAT recorder was interfaced with an Apple Macintosh G5. Altogether, 320 minutes of recordings were analyzed which yielded 2220 segments, 2 formant values/segment, i.e. 4440 measurement points. The current study opts for an interval measurement in the F1 and F2's most stable phases. A PRAAT script automatically measures the labeled vowel formants 50 ms around the middle of the vowel including its standard deviation.⁶ Formant measurements were rechecked by hand and by observing the output standard deviations in *Microsoft Excel*. Fig. 2-1 shows the vowel measurement as applied in this study.

⁵ The DAT recorders were of the types SONY DTC-57 ES, SONY TCD-D7, SONY TCD-D8, and SONY TCD-D100; three microphones of the type SONY ECM-MS907 and a Behringer B2 condenser microphone were used with a directive angle of 90°. Recordings were made with a sampling frequency of 48kHz.

⁶ The script was kindly provided by Beat Siebenhaar from the University of Berne / University of Leipzig.

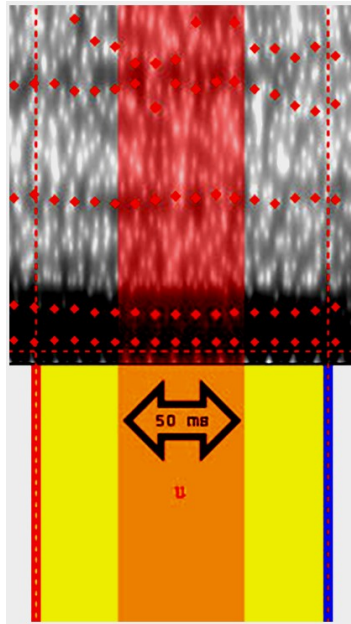


Figure 2-1: Vowel measurement of 50 ms around the nucleus of the vowel including its standard deviation.

In case the vowels could not be measured with the PRAAT script, due to unfavorable spectrograms for example, formants were measured manually according to their linear predictive coding (LPC) traces. Here, the temporal midpoint of the nucleus of the target formant was measured. If LPC traces were completely absent the formants were measured according to their grey scales.

3. Results

During recordings, it was noticed that third of the three attempts was articulated correctly in most cases, while in the first and second attempts, the subjects mispronounced tokens. It is probable that this improvement is due to learning effects. The implications of such mispronunciations are, however, crucial in that the variation between the three attempts might be so significant that subsequent tests with the mean values of the three attempts are disputed. Consequently, it first needed to be tested to what degree such variation exists in order to answer the question if mean values could be used for further analyses.

3.1 Variation between three attempts

A repeated measures ANOVA was run to test for this variation. Results indicated that it is mainly front vowels that present variation between the three attempts; yet, only one p value under the threshold level .05 appeared. Back vowels in contrast, are articulated without far-reaching variation. Vowel plots are helpful to visualize these kinds of findings. Fig. 3-1 shows how the 20 subjects articulated the *ash* vowel (mean of the 3 attempts), which stands in stark contrast to the minor variation of the *foot* vowel (mean of the 3 attempts), see Fig. 3-1.⁷

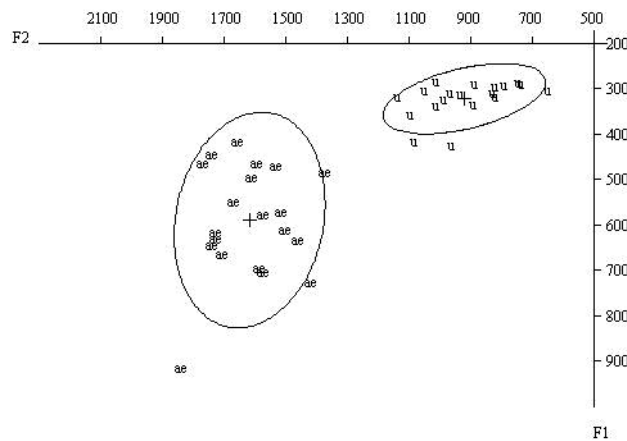


Figure 3-1: Vowel plot of SE /æ/ and /ʊ/.

The high degree of variation on both axes, high-low (F1) and front-back (F2), in /æ/, for instance, can be ascribed to the speakers' varied articulation of the word *had*, carrying the target *ash* vowel, as /hæd/, /həd/, /had/, and /hed/. The ellipse around *ae* symbolizes the sum of variation found in /æ/ - contrasting the small amount of variation we find in /ʊ/, for example. Fortunately, the results of this repeated measures ANOVA showed that it was only the differences between the three attempts in the F1 value of /æ/ which were significant; i.e. 1 out of 20 potentially significant variations. This led to the conclusion that, despite initial reservations as to the reliability of the data set, the data set could be used for further analyses.

⁷ Vowel plot created with *JPlot Formants v.1.4* (Jplot Formants v.1.4)

3.2 Comparison to RP

With t-tests, the SE F1 and F2 values were compared to the values of the RP and GA vowels. A vowel plot best demonstrates how and where the two varieties differ; Fig. 3-2 below shows just that. RP values, retrieved from Deterding's study (1997) are shown in bold black IPA symbols; the SE versions thereof are circled.

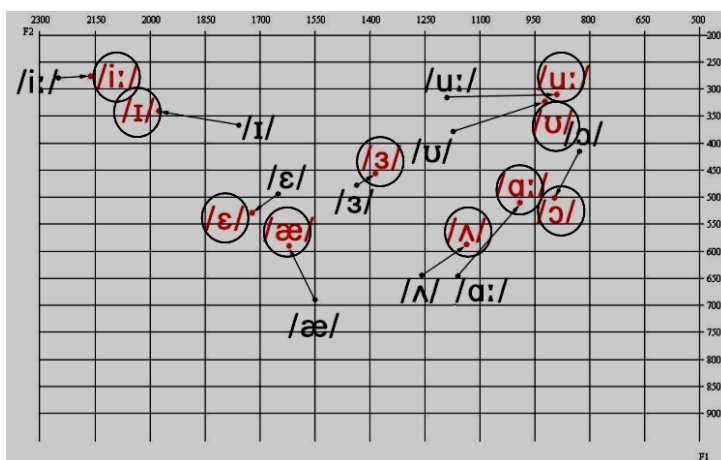


Figure 3-2:
Vowel plot
of SE
(circled) and
RP.

The vowel plot is fairly self-explanatory. At a first glance it is obvious that the SE vowels /ʊ/, /u:/, /ɑ:/, and /ɪ/ are farther away from the target vowel than the SE vowels /i:/, /ε/, /ɜ/, /æ/, /ʌ/. Whether these distances in the vowel space are significant is again tested by means of one-sample t tests for each vowel in its F1 and F2 values.

When looking at the F1 values, the Swiss articulate /ɑ:/, /ʊ/, /ɪ/, /æ/, /ɜ/, /ʌ/ ($p < .05$) and /i:/, /u:/ (not statistically significant) higher than RP speakers do; the vowel, /ε/ and /ɔ/, in contrast, significantly lower ($p < .05$). As regards vowel backness, the F2 values, the Swiss pronounce /i:/, /ɜ/, /ʌ/, /ɑ:/, /ʊ/, and /u:/ extensively further back than RP ($p < .05$) and /ɪ/, /ε/, /æ/ ($p < .05$) and /ɔ/ significantly further fronted ($p < .05$).

3.3 Comparison to GA

The same calculations are then made for SE and GA, plotted in Fig. 3-3.

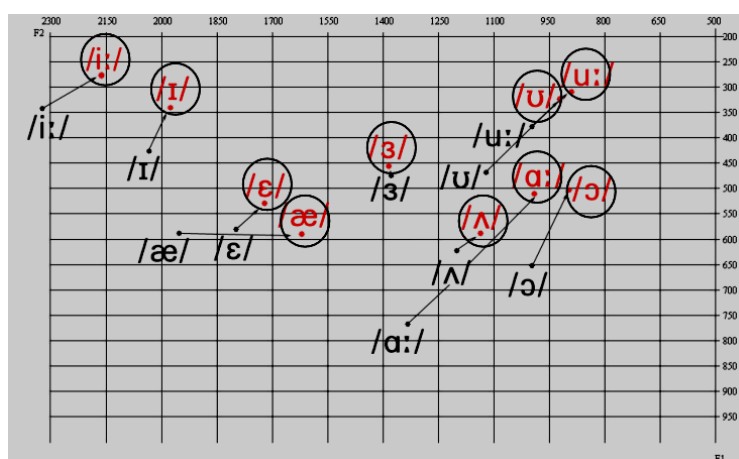


Figure 3-3:
Vowel plot
of SE
(circled)
and GA.

It is clearly visible that /ɛ/, /ɜ/, and /ʌ/ are similar in GA and in SE. The other 7 vowels show greater vowel space distances in either their F1 and/or F2 values. Additionally, it is the vowels /ɑ:/, /i:/, and /ɔ/ which differ most between SE and GA. These first impressions are again corroborated with numbers and figures which show that, with the exception of the *strut* and the *ash* vowel, the Swiss articulate all their SE vowels significantly higher than the speakers of GA do ($p < .05$). The *strut* vowel is articulated only a little higher by the Swiss ($p > .05$) and the *ash* vowel is even a bit lower than the American English /æ/ ($p > .05$). The results are even more unequivocal when we look at the differences in the F2 values between SE and GA: all Swiss English vowels are articulated further back ($p < .05$), except for the centralized retroflex vowel /ɜ/, which is minimally more fronted by the Swiss.

3.4 Synthesis of results

These results need to be considered in greater detail in such a way that it becomes clear, whether the Swiss under scrutiny articulated the target vowels closer to RP or closer to GA. In order to be able to draw such conclusions, each SE vowel and its corresponding p values obtained through the tests between SE & RP and SE & GA is examined. Table 3-1 pools together the mentioned p values. The first column shows the target vowel, the second splits this vowel into F1 and F2, the third column

displays the obtained p value retrieved from the t-test SE & RP, and the fourth column shows the p value obtained from the t-test SE & GA. The last column indicates whether the Swiss pronunciations are closer to RP or GA in that the more the p value approximates zero, the greater the differences in articulation between the groups.

VOWEL	F1/F2	p RP	p GA	Swiss closer to
/i:/	/i:/f1	0.35879	7.02E-14	RP
	/i:/f2	0.00497	1.24E-05	RP
/ɪ/	/ɪ/f1	0.00048	2.08E-11	RP
	/ɪ/f2	1.55E-08	0.01957	GA
/ɛ/	/ɛ/f1	0.05107	0.00703	RP
	/ɛ/f2	0.01046	0.00461	RP
/ɜ/	/ɜ/f1	0.00866	0.02666	GA
	/ɜ/f2	0.00691	0.76001	GA
/æ/	/æ/f1	0.00168	0.92818	GA
	/æ/f2	0.02262	3.31E-10	RP
/ɑ:/	/ɑ:/f1	4.06E-07	1.26E-11	RP
	/ɑ:/f2	3.73E-06	3.78E-11	RP
/ɔ:/	/ɔ:/f1	0.00011	1.55E-07	RP
	/ɔ:/f2	0.01165	0.00106	RP
/ʊ/	/ʊ/f1	8.92E-06	2.19E-13	RP
	/ʊ/f2	2.23E-07	4.95E-06	GA
/ʌ/	/ʌ/f1	0.03607	0.17455	GA
	/ʌ/f2	3.80E-06	0.0028	GA
/u:/	/u:/f1	0.35992	2.70E-08	RP
	/u:/f2	5.99E-09	0.00114	GA

Table 3-1: P values comparison of SE & RP and SE & GA. The last column specifies whether the Swiss are closer to RP or GA.

Table 3-1 explains how /i:/, in both its F1 and F2 values, is articulated more RP-like. In its F1, the front vowel /ɪ/ is also pronounced more like RP, with respect to /ɪ/'s backness, though, the Swiss perform closer to GA. The front vowel /ɛ/, also, sounds more RP-like, yet the p values for RP F1 and F2 are only modestly greater than those of GA. /ɜ/ is articulated more GA-like, especially in its F2 value. In terms of its F1 value, for the low front vowel /æ/ we find a virtually identical articulation as in its GA counterpart. In terms of the F2 value of /æ/, on the contrary, the Swiss are closer to RP. The Swiss' English articulation of /ɑ:/ exhibits the greatest distance from the target varieties. Despite these great discrepancies between SE and GA /ɑ:/ as well as between SE and RP/ɑ:/, the vowel in question is articulated more RP-like in both its frontedness-backness (F2) and its low-height (F1) parameters. The articulation of /ɔ:/ is more comparable to RP than to GA. /ʊ/ illustrates the same p value patterns as /ɑ:/, in that the vowel is pronounced very differently from GA and RP. With regard to the first

formant of /ʊ/, the Swiss approximate RP pronunciation, while, for the second formant, they tend towards a GA articulation. The *strut* vowel shows an Americanized pronunciation in both its F1 and its F2. Finally, the vowel /u:/ sounds more RP-like regarding its height; its frontedness-backness realization is closer to GA.

In sum, then, with respect to vowel height, seven out of ten SE vowels show a more RP-like pronunciation, while three are pronounced rather GA-like. When looking at the vowel backness and frontedness, the results balance themselves out in that the Swiss perform five vowels in a GA fashion and five in RP fashion. It should be pointed out, however, that the interconnectedness between the F1 and F2 values makes the formulation of generalizations difficult. Nevertheless, the present data suggest that, as far as vowels are concerned, the Swiss group tends towards an RP performance rather than towards articulating the vowels in a GA fashion.

The central question remains as to how these findings can be accounted for. The rest of this paper attends to exactly this issue. It tries to detect potential correlations between external variables, elicited with the questionnaire, and the students' SE performance.

4. Seeking explanations

Causes for variation in second language production are psychologically and socially complex in nature – which is why the questionnaire tried to elicit as many external factors as possible. The questionnaire, which the students filled out after the recordings, was divided into 3 sections: demographics, L2 background, and language attitudes. With regard to demographics, the subjects were asked to indicate sex and age. Neither of these variables had a significant impact on their L2 articulation; 20 year-olds did not articulate SE vowels differently from 16 year-olds, for example. In terms of L2 background and language attitudes, however, significant differences between the L2 learners of English were found – two findings which will be addressed in greater detail in this final section.

4.1 Active pronunciation training

In order to find out more about the students' L2 background, they provided information as to how often they speak English, read English, listen to English, if and how long they have been in an English-speaking country (cf. Flege and Liu 2001, for example), about the target variety taught in their English classroom, and whether they receive active pronunciation training in class (cf. Catford and Pisoni 1970; Matthews 1997). All but one of these variables did not provide significant

results. It was only in the last aspect that was mentioned, active pronunciation training, for which p values under .05 were obtained.

As early as in 1970, Catford and Pisoni (1970) found proof that students who receive systematic training in articulatory phonetics perform closer to target forms than those who are not exposed to such training (a finding which was later substantiated by Matthews (1997) for example). For this reason, the students were asked whether their English teachers principally focus on lexis, syntax, pronunciation, or interactional competence. 11 out of 20 students say their teachers particularly focus on articulation. An independent-samples t test shows that there are significant differences between the SE vowel quality of those speakers who did have active training and those who did not. We find significant differences between the two groups in the F1 values of /ʊ/ and /u:/ (p < .05). Zooming in on the results, tests show that students who were instructed in pronunciation demonstrate 342 Hz in the F1 for /ʊ/, as opposed to 303 Hz for those who did not undergo articulation training. Deterding's (1997) and Hillenbrand et al.'s (1995) values suggest an F1 value for /ʊ/ RP of 379 Hz and an F1 value for /ʊ/ GA of 469 Hz – these numbers suggest that those students who were taught pronunciation in effect are closer to the standard varieties than those who were not instructed. A matching pattern is found in the realization of the /u:/ vowel.

In spite of the fact that only two out of 20 p values are significant, these results bear a comparatively high explanatory power as they rely on almost evenly distributed group sizes. Additionally, I venture to suggest that it is not a coincidence that the vowels /ʊ/ and /u:/ emerge as points of distinction between the two groups. In many English classrooms in Switzerland where teachers are concerned about their students' pronunciation, practice in the discrimination between these two vowels is a frequently revisited issue.

4.2 Subject preference for RP or GA

Since the emergence of language attitude studies in the 1960s, it is widely recognized that people show entrenched attitudes towards speakers of different speaking styles (cf. Giles et al. 1987). Therefore, the subjects were asked to provide information as to their attitudes towards RP and GA, which included preference for RP or GA, subjective perception of “nice” English, level of embarrassment when speaking English, finding pleasure in speaking English, and wanting to attain the same pronunciation as their English teacher. The only significant results with respect to these variables were obtained by testing a correlation between the subjects' preference for RP and GA and their articulation of Swiss English.

Eight of the 20 subjects prefer GA and twelve favor RP. The independent-samples t test proposes an intriguing picture: The SE F1 values of /æ/, /ɔ:/, /i:/, /ɑ:/, and /u:/ between those who prefer RP and those who like GA better are significantly different ($p < .05$).

Again, let us look at the results in greater detail. Descriptives show that the mean F1 value of /æ/ is 514 Hz for those who prefer GA and 641 Hz on average for students who claim to favor RP. The F1 values of /æ/ RP and /æ/ GA are 690 Hz for RP (Deterding 1997) and 588 for GA (Hillenbrand et al. 1995). With regard to /æ/ F1, one can infer that those who favor RP indeed perform closer RP, and those who say they favor GA speak more GA-like. As regards the F1 value of /ɔ:/, descriptives show that speakers who favor American English present 453 Hz whereas those who have an inclination for the British variety display 536 Hz. The F1 values for /ɔ:/ in the standard varieties are 415 Hz in RP and 652 Hz in GA. In this case we find a contradiction to the previous finding: those who say they prefer GA are in fact closer to RP and vice versa. As regards the F1 of /i:/, the group that likes GA better shows 285 Hz as opposed to those favoring RP, showing 270 Hz. The values of the Standard Englishes are 342 Hz in GA and 280 Hz in RP. Even though the difference in Hertz in the F1 value of /i:/ between the subjects who prefer GA and the subjects who like RP only makes up 10 Hz, one can again argue that the subjects who state preference for RP perform closer to their target variety RP, and those who find more enjoyment in American English verbalize more GA-like vowels. However, we find another counter-intuitive example in the vowel /ɑ:/: the Swiss who prefer GA show 461 Hz and those who favor RP show 543 Hz – while the Standard English values of /ɑ:/ F1 is 646 Hz in RP and 768 Hz in GA. Again a negative correlation between what the students say they like and how they perform their target English vowels is found. The last value to be considered is the F1 of /u:/. Subjects who like GA show 332 Hz and those who prefer British English display 298 Hz. The mean values of the /u:/ F1s for the Standard Englishes are 316 for RP and 378 for GA. Another case where those who prefer GA are closer to GA and similarly, those who prefer RP speak more RP-like.

These findings suggest that preference for RP or GA shapes English articulation. If more subjects had been scrutinized, such arguments could probably be substantiated. These results, in turn, contradict Lasagabaster et al.'s (2004, 401) claim that “there seems to be a gap between what people say (their expressed attitudes) and what they do (their actual behavior),” as the Swiss under analysis in effect seem to form their actual verbal behavior in line with their preference for GA or RP.

5. Conclusion

This paper scrutinized the English vowel quality of 20 Swiss German EFL speakers. The study has shown that virtually all SE vowels are pronounced significantly different from their RP and GA targets. In contrast to RP, the Swiss articulated /ɑ:/, /ʊ/, /ɪ/, /æ/, /ɜ/, /ʌ/, /i:/ /u:/ (the last two not statistically significant) significantly higher, whereas the vowels /ɛ/ and /ɔ/ were pronounced significantly lower in Swiss English. In terms of backness and frontedness, the Swiss articulated /i:/, /ɜ/, /ʌ/, /ɑ:/, /ʊ/, and /u:/ significantly backened and /ɪ/, /ɛ/, /æ/, and /ɔ/ significantly more fronted than in RP. In contrast to GA, the SE variety shows significantly higher vowels than GA, except for /æ/ (not statistically significant) which was pronounced somewhat lower. As for the F2 values, the subjects articulated each vowel significantly further back, except for /ɜ/ (not statistically significant), which was minimally more fronted. Thus, there seems to be an overall trend of the Swiss group to pronounce English vowels more RP-like than GA-like, specifically in terms of the vowels' F2 values.

The second half of the paper was devoted to seek explanations for these findings. Statistical tests, were run to examine whether SE means correlated with variables collected with the questionnaire. Two interesting findings stand in the foreground. One is that students who were given pronunciation training articulate the F1s of /u:/ and /ʊ/ more native-like than those who were not exposed to articulatory training. Secondly, there is statistically reliable evidence that the subjects' preference of RP over GA and vice versa exerts influence on their SE vowel quality. It has been pointed out that, with respect to a number of vowel values, students who favor GA perform more GA-like, while subjects who favor RP articulate certain vowels more RP-like.

This study has, if only to some extent, corroborated the assumptions that L2 articulation is affected by attitudes towards the target language and that L2 pronunciation instruction may alleviate a foreign accent. From my personal experience I know that, most of the time, foreign language teachers in Switzerland do not place particular emphasis on pronunciation training – possibly because there may be a lack of knowledge to do so. Thus, if teachers want to assist their learners in reducing their Swiss-German flavored accents of English, teachers themselves must be in the know about basic articulatory phonetics. What is more, however, is that teachers need didactically useful strategies to communicate this knowledge to their students – that is only, of course, if the students and if the teachers deem a foreign accent undesirable.

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